INTRODUCTION TO MODEL CATEGORIES

INCOMPLETE

Karshit Joshi
Math and Physics Student
Chennai Mathematical Institute
9th September 2026

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1 Preliminaries

This is an exposition written by me to take notes of what I read about Model Categories and certain ideas that I think are very useful while reading Model Categories.Any mistakes made in this exposition is purely mine and please report them to [karshitkjoshi@cmi.ac.in].I am mostly using "Homotopical Algebra" by Daniel Quillen[1],"Model Categories" by Mark Hovey[2] and "Categorical Homotopy Theory" by Emily Riehl[3] as my primary resources to read about the topic from, and will try my best to mention any other resources that I use along the way in the bibliography.(11th October 2026) I have decided that I shall write about infinity categories and its relation to topological spaces along with model categories and also focus on quasi-category model of infinity category, for which my primary resource is this playlist of this course[4] and "A short course on Infinity Categories" by Moritz Groth[5].I am grateful to Chennai Mathematical Institute to expose me to the resources and wonderful people that helped me during this writing

I am very grateful to Prof. Aditya Karnataki, who encouraged me to write this exposition and introducing me to writing exposition which led me to find an appreciation for writing mathematical expositions.

2 Derived Functors

2.1 Homotopical Categories and Derived Functors

An important question has always been in different settings and even in category theory, "Are these two objects isomorphic ?", while the answer to the question depends on the what we consider an isomorphism.Fixing the context, it could perhaps be easier or difficult to differentiate between objects.

A good strategy in our toolset has always been to classify objects "upto some property".Note that the creation of such classification creates a new category where the context is changed to something which is much coarser than the current context that we are trying to work in.This often helps us to focus ourselves towards a particular way to differentiate objects rather than having the vague goal of differentiating objects with nothing ahead of us.Common examples of this phenomenon include Homotopy Groups, Homology Groups, Euler Characteristic, Betti Numbers, etc.We shall now try to build this notion in category theory for abstract categories.

Suppose we start with a category $\mathcal{C}$.We have some property $\mathcal{P}$ along which we can test our objects to see if they are "equivalent" or not.If identifying $\mathcal{P}$ on 2 different objects tells us that they have the same property.Our desire is to consider those objects to be the same.A natural way to store all of this data of identification is in a category, with the purpose that we will inherit all the data naturally by doing a construction on $\mathcal{C}$.

We are now clear that we must perform construction on $\mathcal{C}$ to create a category which characterizes the data that we get from $\mathcal{P}$.There are naturally 2 ways to do this.[1]

- Consider the objects of $\mathcal{C}$, consider the equivalence relation imposed by $\mathcal{P}$ on the collection of object of C. We quotient by the equivalence relation. We also forget all the maps between objects which were identified.The residue will turn out to be a category with a few more natural steps. This can be thought of as skeleton category of localisation given by the second point.

- Consider the maps of $\mathcal{C}$, formally introduce the inverses of maps between objects, "which characterize the equivalence of $\mathcal{P}$".This just naturally forms a category once again after completing all possible compositions with the new maps.[2]

Both the above ways are useful in different ways that we will explore throughout the span of this text. Notice the difference, that the first construction forgets how things are actually equivalent whereas the second does not due to the existence of all the maps which "characterize the equivalence of $\mathcal{P}$.This especially comes into the picture if you view the first construction as the manifestation of "existence of an equivalence" rather than "equivalent

[1] A person who already has a vague idea about infinity categories might realize that the 2 above localisations can be unified under a kind of construction that can be done in infinity categories called n-truncation. This allows to truncate all data from n and above to get rid of the precise witnesses which exhibit the homotopy between (n-1) morphisms and replace them with just existence statements

[2] There is an interesting question for keen observers. There are 2 ways to achieve this. (1) Introduce the "abstract" formal inverses of all maps we consider weak equivalences (2) Create an equivalence relation between maps and quotient by the equivalence relation which will force certain "concrete" maps to become formal inverses of other "concrete" maps. When exactly are these constructions equivalent/isomorphic ? In some sense, the answer for why they would be equivalent is that having a left inverse upto some homotopy can make the inverse unique upto homotopy which we can test if all such inverses have a characterisation in the base category or not. I still need to know more about this though.

with the witness map", wherein the second construction is much more suited to the latter. This often comes into play when you are considering n-truncations of an infinity category which are the objects we are truly trying to study indirectly via model categories. Due to the previous reason, this also plays a significant role in homotopy type theory.[Note that the first construction is also technically inline with Leibniz equivalence of identibles which says that "if two objects have the same interactions, they are the same".i.e. Isomorphisms should not exist as objects, instead all isomorphisms must be equalities. This plays a major role in homotopy type theory.]

Since the second construction retains much more information from the ambient category. It is natural to call it the appropriate "localisation" of the category. The terminology of localisation is inherited from ring theory, wherein we localize a ring against a prime ideal which requires to invert all elements outside the prime ideal.

We now turn our focus to actually selecting the maps of the category $\mathcal{C}$, to be our equivalences which "we believe" characterizes some property $\mathcal{P}$. Here is a subtle but important point that we "choose" a property to classify objects against. In the context of abstract categories, the selection of maps in the category is in direct correspondence to the abstract property $\mathcal{P}$. Hence, the selection of property is equivalent to selection of which maps to invert. Hence, the structure we obtain is a deliberate construction, not a natural structure to consider, i.e. The structure is a choice, not a natural byproduct of $\mathcal{C}$.

It is also natural to expect that all isomorphisms are also a part of the collection of maps we invert mainly because, if $\mathcal{C}$ cannot distinguish between 2 objects, they must possess the same set of properties at all levels [Note that this is essentially the application of Yoneda Lemma]. Hence, this helps us to arrive at our first definition.

Definition 2.1. A **homotopical category** is a category $\mathcal{C}$ equipped with a wide* subcategory $\mathcal{W}$ such that, for any composable triple of arrows,i.e. f, g, h. If hg and gf are in $\mathcal{W}$, so are f, g, h, and hgf.
The arrows in $\mathcal{W}$ are called equivalences. The property above concerning the maps is called the 2-of-6 property.[abbrievation of "2 out of 6"] (*) By a wide subcategory, we mean a subcategory which contains all identities of the category $\mathcal{C}$ and hence by 2-of-6 property, all isomorphisms of $\mathcal{C}$ aswell.

There is a weaker condition that is also often considered instead of the 2-of-6 property. It is called 2-of-3 property.

Definition 2.2. 2-of-3 Property occurs if any 2 of the 3 possible composition of 2 composable maps are in $\mathcal{W}$. Then $\mathcal{W}$ contains all 3.

A small caveat in the above 2 definitions is that $\mathcal{W}$ must also contain all the identities for each object in $\mathcal{C}$ which has a clear enough reason to be included.

Now that we have concretely defined what is the data that we would like to work with. We now approach the problem of carrying out the construction of the localisation of the category as we previously mentioned. The following explicit construction is due to Pierre Gabriel and Michel Zisman. Its morphisms are equivalence classes of finite zig-zags (i.e.compositions which maps from $\mathcal{W}$ in the string with finite length) with arrows in $\mathcal{W}$ to be inverted with

the same name but dual of its original counterpart subjected to the following relations.

Definition 2.3. The localisation category/homotopy category Ho$\mathcal{C}$ of a homotopical category $(\mathcal{C}, \mathcal{W})$ is the formal localisation of $\mathcal{C}$ at the subcategory $\mathcal{W}$. The explicit construction of the homotopy category is as follows:

- Objects of Ho$\mathcal{C}$ are the same as $\mathcal{C}$.
- All strings of compositions which contain $\xleftarrow{w}\xrightarrow{w}$ or $\xrightarrow{w}\xleftarrow{w}$ or identities pointing forward or backwards as a part of the string are equivalent to strings of compositions with the earlier substrings removed. This induces the natural equivalence relation of strings of composable arrows.

One can simply notice that there is a natural functor $\mathcal{C} \xrightarrow{L}$ Ho$\mathcal{C}$. This functor also possesses the universal property that any other functor which inverts all morphisms in $\mathcal{W}$ (not necessarily only morphisms in $\mathcal{W}$) factors through L.

An important point is that not all morphisms of $\mathcal{C}$ that become isomorphisms should be present in the subcategory $\mathcal{W}$. When this does not happen, we call the homotopical category $\mathcal{C}$ to be saturated [i.e. all morphisms whose inverse exists in Ho$\mathcal{C}$ are also present in $\mathcal{W}$].

Another important point is that performing this construction on a locally small category does not ensure that it will remain a locally small category. This is simply due to the reason that the collection of objects form a class and each object has its respective equivalence class. There could be a lot of choices of a single map about which object to passthrough for each object's equivalence class. Considering the possibility that the equivalence class of an object may itself form a class. The collection of maps between any two maps can form a class. For this reason, there exists another richer localisation than the one we came up with, which is inspired from the localisation of rings, called "Calculus of Fractions" due to Pierre Gabriel, Pierre Zisman. However, I plan to delve into it just yet.

Therefore, one of the problems that the model category construction solves is to keep the localisation of the category to be locally small.

We now try to study functors between Homotopical Categories which can extend to maps between the respective homotopy categories under suitable conditions on the functor itself. The condition on the functor is also natural, in the sense that it must send equivalences to equivalences ,i.e. it preserves all equivalences. We call such functors, "Homotopical Functors", by the universal property of localisation, it induces a functor between the respective homotopy categories.

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{\quad F \quad} & \mathcal{D} \\ \downarrow{\scriptstyle L_{\mathcal{C}}} & & \downarrow{\scriptstyle L_{\mathcal{D}}} \\ Ho\mathcal{C} & \xrightarrow{\quad F' \quad} & Ho\mathcal{D} \end{array}$$

It is a natural desire to study functors which are not homotopical. However, when that happens notice that $L_{\mathcal{C}}$ and $L_{\mathcal{D}} \circ F$ form a span diagram, which hints us towards looking at left and right Kan Extensions along the appropriate direction carrying the intuition that we are trying to find the best functor which approximates F while losing the least amount

of information while doing it. This naturally leads to the following definition.

Definition 2.4. A **total left derived functor** LF of a functor F between Homotopical Categories $\mathcal{C}$ and $\mathcal{D}$ is a right Kan extension $Ran_{L_\mathcal{C}}(L_\mathcal{D} \circ \mathcal{F})$ [We shall also sometimes refer the $Ran_{L_\mathcal{C}}(L_\mathcal{D} \circ \mathcal{F}) \circ L_\mathcal{C}$ as the total left derived functor with an equivalent universal property]

This also alludes to being related to derived functors in the homological algebra sense. We are eventually heading towards that direction. Now we introduce the definition of a left derived functor which won't just yet make any sense but its better to go through the definition first and then realise it.

Definition 2.5. A **left derived functor** of $F : \mathcal{C} \to \mathcal{D}$ is a homotopical functor $\mathcal{L}F : \mathcal{M} \to \mathcal{N}$ along with a natural transformation $\tau : \mathcal{L}F \Rightarrow F$ such that $L_\mathcal{D}\tau : \tau \circ \mathcal{L}F \Rightarrow L_\mathcal{D} \circ F$. the last operation is also better known as "whiskering" ,i.e. left derived functor is a direct lift of the total left derived functor along the localisation functor.

The total left derived functor is essentially nothing but the induced functor between the homotopy categories from the functor between the homotopical categories. In classical notion, this will only occur when the functor respects the weak equivalences that we have chosen ,i.e. the functor is well-defined on the isomorphism classes of objects equivalently F is not a homotopy functor. The main problem is that most functors that we are aware of ,i.e. $Hom(-, A)$, $- \otimes A$ do not respect the quasi-isomorphisms in the category chain complexes. This tells you that a total left derived functor. Now, borrowing the intuition for a Kan Extension, One can realise that this is precisely the problem that we would like to solve which is why the goal is to find the total left derived functor ,i.e. the best approximation of F. We now introduce some more definitions which are directly motivated from the topology.

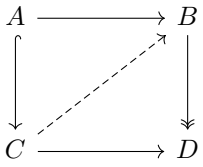
Definition 2.6. A **left deformation** on a homotopical category $\mathcal{C}$ consists of an endofunctor $\mathcal{Q}$ together with a natural "weak equivalence", not natural isomorphism instead its a natural isomorphism as a homotopical endofunctor on the homotopy category $q : \mathcal{Q} \Leftrightarrow 1$.

We are already tip-toeing into the deep waters of model categories, as we go into the following example which considers the weak factorisation system. If $\mathcal{M}$ is a model category with a functorial cofibration-trivial fibration factorisation, then the action of this factorisation on arrows of the form $\phi \to x$ produces a functor $\mathcal{Q}$ whose image lands in the subcategory of cofibrant objects with natural weak equivalence $q_x : Qx \sim x$. I just copied the previous example verbatim from Riehl's book but I believe this is an important example to look at.

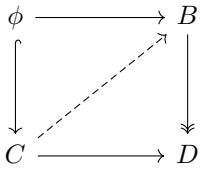
In the above example, we use a weak factorisation system to break any map into a cofibration and weakly equivalent fibration which allows us to find an object q_x for each x for which $\phi \to x$ is a cofibration and it is equivalent to x which will constitute a larger functor $q : x \sim q_x$ sending the category to its subcategory of cofibrant objects/replacements. As it will soon turn out that such replacements are of extreme importance to create what

are homotopy limit/colimits which are grave importance to work in the homotopy category, which in most cases does not even have all limits and colimits. A reasonable question to ask is that why q_x is called a cofibrant object/replacement even though its weak equivalence with x is a fibration. There is a very important reasoning behind this. It is for, How exactly do we deal with extension/lifting problems ?

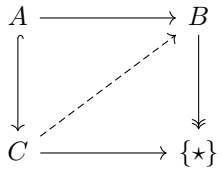
Many times in mathematics, again and once again you must encounter this precise diagram.



When does exactly such a lift/extension [dotted arrow] exist such that both the triangles commute ? Often times, a lifting or extension problem can be precisely phrased in this way given the existence of an initial and a terminal object.



Lifting Problem



Extension Problem

And it is due to the prominent of these problems which are solved by cofibrant objects because the property of $\phi \to C$ being a cofibration is precisely that it lifts all maps against any fibration $B \to D$. This solves the lifting problem for any object C because as long as $B \to D$ is a fibration, in the homotopy category I can replace C with q_C and lift any map $q_C \to D$ without any worries. A dual situation will occur when weak factorisation system is trivial-cofibrations and fibrations. Wherein you will be replacing objects with fibrant replacements. This is precisely why a Weak Factorisation System in ingrained into the definition of a Model Category because this allows us to have some level of control over the maps in the homotopy category [while obviously increasing a lot of conditions on the original category like i.e. there existing a weak factorisation system in the first place] and evidently enough, this enough for us to get limits and colimits of diagrams in the homotopy category which inturn allows us to actually compute legible objects which are significant to us. But given a significant idea of how we are going to solve our problems, we leave this here for the reader to digest this situation as this is one of the major parts of model categories which build the bigger picture.Instead we try to bring back our focus onto realising the

current definitions with our knowledge from Homological Algebra.[3]

We know that the category of left R-modules have the following property: for nay R-module A, there exists a projective module P together with an epimorphism $P \twoheadrightarrow A$. This way, repeating the procedure we get the **projective resolution**, an acyclic (except in degree zero) bounded-below chain complex of projectives that maps quasi-isomorphically to the chain complex concentrated in degree 0 at module A. Making a specific natural choice of projective resolution (free resolution), gives us a left deformation for the homotopical category of tbe homotopical category of bounded-below chain complexes where the weak equivalences are the quasi-isomorphisms.Dually, any R-module embeds into an injective module, repeating that gives us injective resolutions. Similarly, one can arrive at right deformation on the category of bounded-below cochain complexes.

Any additive functor F from the category of R-modules to category of S-modules induces obvious functors between the respective chain complex categories. These functors preserve the chain homotopy equivalences by very easy verification.But, they do not preserve quasi-isomorphisms which are the actual weak-equivalences that we would like to look at because they are completely faithful to homology/cohomology groups. Although, for projective resolutions, any chain complex made of projective objects, a quasi-isomorphism entails a chain homotopy equivalence which is preserved by $F_\bullet$. The dual versions of the above statements also follows correctly. This also fits into the previous picture that we are essentially taking a cofibrant replacement in the category of chain complexes and since F preserves the quasi-isomorphisms in the subcategory of projective chain complexes. We can get a total derived functor between the respective homotopy category chain complexes.

The classically regarded as the left derived functor of $F : \mathbf{Mod}_R \to \mathbf{Mod}_S$ can now be viewed as below using what we have realised up until now:

$$\mathbf{Mod}_R \xrightarrow{deg_0} \mathbf{Ch}_{\geq 0}(R) \to \mathbf{Ch}_{\geq 0}(S) \xrightarrow{H_0} \mathbf{Mod}_S$$

which takes the cofibrant replacement of an R-module A, applies $F_\bullet$, and then computes the 0th homology. Please note that it is very possible that the natural deformation functors does not exist for your abelian category, in which case also this procedure will pass through just by knowing that there is a weak factorisation system on the category. Also note that if you do not have such a natural deformation functor then you will not get a derived functor but instead a total derived functor due to the reason that a person now must make a choice of cofibrant replacement for any object rather than there being a natural choice to pick for each choice.

At this point I expect the reader to understand that why are does one care about fibrations, cofibrations, weak equivalences, weak factorisation systems, fibrant and cofibrant replacements, which are some of the most important objects in understanding/dealing with model categories. Now, it is an adequate point to take a look into homotopy (co)limits and how are they exactly constructed while expecting the reader to have realised the importance of knowing how to reliably construct limits/colimits in the homotopy category and why will they be very useful. Now we shall go into a bit of enriched category theory as it is useful in model categories framework, calculating homotopy (co)limits aswell as understanding models of infinity categories.

[3]I insist the reader to atleast read the definition of a Weak Factorisation System at this point in time.

2.2 Enriched Category Theory

The core idea behind enriched category theory is that to try to localise a category, you must invert maps which in turn force you to deal with the hom-sets or in many cases hom-objects which now have more structure because our desire to localise. Therefore, to keep track of data as it behaves in the ambient category but also be able to see the consequences of localisation as play around with hom-objects which is precisely what enriching a category is all about, replacing hom-sets with hom-objects in some other category D.

Precisely speaking, we are going make our hom-objects precisely to be spaces/topological spaces which carry the information of homotopies and higher homotopies in the terms of paths, paths between paths, paths between paths between paths, ... up until infinity (i.e. simplicial set, which can also be realised to be a CW-complex).

Definition 2.7. The base category over which we enrich must be a **symmetric monoidal category** $(V, \times, \star)$. Here V is an ordinary category. $\times : V \times V \to V$ is a bifunctor and $\star$ is the unit object. It is also equipped with the following natural transformations.

$$v \times w \cong w \times v \ , \ u \times (v \times w) \cong (u \times v) \times w \ , \ \star \times v \cong v \cong v \times \star$$

We also do not have to worry about the associativity holding at higher levels.

The monoidal product is supposed to serve the composition operation as well as the unit is supposed to be the initial object in the category V to point out the identity map for each endo-object for each object.

Definition 2.8. A V-category $\underline{\mathcal{D}}$ consists of

- a collection of objects, i.e. x,y,z $\in \underline{\mathcal{D}}$
- for each pair x,y $\in \underline{\mathcal{D}}$, a hom-object $\underline{\mathcal{D}}(x, y) \in V$.
- for each x $\in \underline{\mathcal{D}}$, a morphism $id_x : \star \to \underline{\mathcal{D}}(x, x) \in V$.
- for each triple x,y,z $\in \underline{\mathcal{D}}$, a morphism $\circ : \underline{\mathcal{D}}(y, z) \times \underline{\mathcal{D}}(x, y) \to \underline{\mathcal{D}}(x, z)$ in V.

such that the following diagrams commute for all $x, y, z, w \in \underline{\mathcal{D}}$:

$$\begin{array}{ccc} \underline{\mathcal{D}}(zw,y) \times \underline{\mathcal{D}}(y,z) \times \underline{\mathcal{D}}(x,y) & \xrightarrow{1\times\circ} & \underline{\mathcal{D}}(z,w) \times \underline{\mathcal{D}}(x,z) \\ \downarrow{\scriptstyle \circ\times 1} & & \downarrow{\scriptstyle \circ} \\ \underline{\mathcal{D}}(y,w) \times \underline{\mathcal{D}}(x,y) & \xrightarrow[\circ]{} & \underline{\mathcal{D}}(x,w) \end{array}$$

$$\begin{array}{ccc} \underline{\mathcal{D}}(x,y) \times \star & \xrightarrow{1\times id_x} & \underline{\mathcal{D}}(x,y) \times \underline{\mathcal{D}}(x,x) \\ & {\scriptstyle\cong}\searrow \quad \swarrow{\scriptstyle\circ} & \\ & \underline{\mathcal{D}}(x,y) & \end{array}$$

$$\begin{array}{ccc} \underline{\mathcal{D}}(x,y) \times \star & \xrightarrow{id_y\times 1} & \underline{\mathcal{D}}(y,y) \times \underline{\mathcal{D}}(x,y) \\ & {\scriptstyle\cong}\searrow \quad \swarrow{\scriptstyle\circ} & \\ & \underline{\mathcal{D}}(x,y) & \end{array}$$

The goal here is to take as much advantage of the base category structure as possible. And it turns out that in many nice categories, currying is possible and there is a parametrized adjunction between the $\times$ and the "internal hom". This alludes to the fact that such an adjunction is only possible in a category when the hom-objects of the base category lie within itself ,i.e. the base category is enriched over itself. This is a necessary but also sufficient condition for currying. We obviously cannot let go of such a nice structure, therefore we are going to introduce a definition for it.

Definition 2.9. We call a monoidal category **closed** if each functor $- \times d : \mathcal{D} \to \mathcal{D}$ admits a right adjoint $\underline{\mathcal{D}}(d, -)$, the right adjoints in this family of parameterized adjunctions assemble in a unique way into a bifunctor.

$$\underline{\mathcal{D}}(-,-) : \mathcal{D}^{op} \times \mathcal{D} \to \mathcal{D}$$

such that there exists natural isomorphisms for all $u, v, w \in \mathcal{V}$

$$\mathcal{D}(u \times v, w) \cong \mathcal{D}(u, \underline{\mathcal{D}}(v, w))$$

what follows are the categories which are also called **closed symmetric monoidal categories**, which are natural to understand by now. In the special case that the monoidal product is the cartesian product(i.e. the categorical product) on $\mathcal{V}$, one says that $\mathcal{V}$ is **cartesian closed**. We now introduce a construction which is very useful later on in topology. For any cartesian closed symmetric monoidal category $(\mathcal{V}, \times, \star)$, which we also suppose to be complete and cocomplete. Consider the slice category under $\star$, the terminal object of $\mathcal{V}$. It should be instantly clear what the construction is supposed to be. Let this category be $\mathcal{V}_\star$, otherwise known as the category of **based objects** in $\mathcal{V}$. We define the product to be the **smash product** which is the pushout square of the following kind.

$$\begin{array}{ccc} v \sqcup w & \longrightarrow & v \times w \\ \downarrow & & \downarrow \\ \star & \longrightarrow & v \wedge w \end{array}$$

This is the natural way to define a product on this particular slice category because here we are just quotienting the edges along $\star$ in the product to a single point which can be viewed using the pushout formula or actually calculating the colimit of the diagram. This construction is associative upto isomorphism (it would be associative if the underlying monoidal category is strictly associative). Since $\mathcal{V}$ is cartesian closed, the functors $v \times -$ preserve colimits. This may allow you to consider based spaces where you might not have looked for before. At this point the point of enriched categories should be clear. We would like to makes paths in the mapping space of two objects to create homotopies and create paths between paths to create higher homotopies. This should now be clear that for homotopy theory we would like to enrich our categories over some kind of simplicial objects, perhaps simplicial sets. Although before moving into simplicial categories we shall take a look at something else before.

There is a natural way to define a "good" functor between symmetric monoidal category, i.e. it preserves the symmetric monoidal structure in the sense that we describe below.

Definition 2.10. A functor $F : \mathcal{V} \to \mathcal{U}$ between symmetric monoidal categories $(\mathcal{V}, \times, \star)$ and $(\mathcal{U}, \otimes, \mathbb{I})$ is **lax monoidal** if there exist associative and unital natural transformations

$$\mathbb{I} \to F(\star) \text{ and } Fv \otimes Fv' \to F(v \times v')$$

One would have expected that the above natural transformation is a natural isomorphism instead. While that is a completely valid way to think, infact there is indeed a definition specifically for them aswell but the above definition also serves the purpose of change of base functors which are of great importance while learning enriched category theory. We shall mention the other variants of the definition down below.

- We call a functor **strong monoidal** if the above natural transformations become natural isomorphisms.
- We call a functor **strict monoidal** if the above natural transformations just become equalities or identities.
- We call a functor **oplax monoidal** if the direction of natural transformations in the definition of lax monoidal functors is reversed.

We now delve into the usage of lax monoidal functors that we mentioned before, Change of Base functors.

Proposition 2.11. In the presence of any lax monoidal functor $F : \mathcal{V} \to \mathcal{U}$, any $\mathcal{V} - category$ has an associated $\mathcal{U} - category$ with the same objects.

The proof really just follows simply from trying to find the object in $\mathcal{U}$ for any string of operations in $\mathcal{V}$ using the natural transformations given by the lax monoidal functor which essentially completes the proof by using the coherence conditions on $\mathcal{V}$ first and then $\mathcal{U}$. This can be used to lose the enrichment structure of any $\mathcal{V} - category$, i.e. change the base of the category from $\mathcal{V}$ to **Set**. Using the proposition for this specific case, we get what is called "losing" the enrichment for an enriched category. The lax monoidal functor that we use to achieve this is the $\mathcal{V}(\star, -)$.

Definition 2.12. The **underlying category** $\mathcal{C}_0$ of a $\mathcal{V} - category$ $\underline{\mathcal{C}}$ has the same objects as before but has the hom-sets as

$$\mathcal{C}_0(x, y) = \mathcal{V}(\star, \underline{\mathcal{C}}(x, y))$$

One can now reasonably think about closed monoidal categories, when you lose the enrichment on a self-enriched category, you would just get the same category back with hom-sets instead of hom-objects assuming the category was locally-small. We shall now talk about enriched functors and enriched natural transformations a little bit before going into the meat of the situation that is the simplicial category.

Definition 2.13. A $\mathcal{V}$-functor $F : \underline{\mathcal{C}} \to \underline{\mathcal{D}}$ btween $\mathcal{V}$-categories is given by an object map between the enriched categories and a map between the base categories of the following form

$$\underline{\mathcal{C}}(x, y) \xrightarrow{F_{x,y}} \underline{\mathcal{D}}(Fx, Fy)$$

I hope this gives you the suitable style in which enriched category theory is defined. You can go on to define enriched natural transformations, enriched Yoneda lemma, enriched adjunctions and so on but we shall stop here and move onto a much more important topic of Simplicial Categories ,i.e. categories enriched over $(\mathbf{SSet}, \times, \star)$.

Any simplicial category $\underline{\mathcal{C}}$ gives a simplicial object in **Cat**. Any simplicial category gives us a simplicial object in the **Cat**, i.e. the enrichment yields a structure such that, therefore at each level you can define a C_n which contains the same objects as in the unenriched category and all n-simplicies(string of arrows with n maps) morphisms between the objects. We may denote this simplicial object as $\mathcal{C}_\bullet$. Under this understanding, one can except that C_0 is the underlying category itself. [I believe theres more to this, hence I must look into it]

Now, we shall try to abstract out a very important concept or rather view it as "getting as close to a self-enriched category" as possible without being self-enriched, in the following sense, and the concrete characterisation of this phenomenon can also be observed in Hom-Tensor Adjunction, which is precisely where it derives its name for.

The goal of getting a $\mathcal{V}$ enriched category $\underline{\mathcal{M}}$ very close to enriching over itself means that the functor $\underline{\mathcal{M}}(-,-)$ needs to have an "approximation" in $\underline{\mathcal{M}}$. This precise statement reminds us of adjunctions which are precisely how tensored categories are defined for us, which is in the following way.

Definition 2.14. A $\mathcal{V}$-category $\underline{\mathcal{M}}$ is **tensored**[a] if for each $v \in \mathcal{V}$ and $m \in \mathcal{M}$, there is an object $v \otimes m \in \mathcal{M}$ with natural isomorphisms,

$$\underline{\mathcal{M}}(v \otimes m, n) = \underline{\mathcal{V}}(v, \underline{\mathcal{M}}(m, n))$$

[a] there is no universally accepted definition of tensored categories, I am just presenting the one given in Riehl's Book.

This is nothing but essentially just the left adjoint of the functor $\underline{\mathcal{M}}(m, -)$, which precisely explains why one would try to think of it in the way that I have mentioned above. Similarly, a cotensored category has a similar feature except it is the right adjoint of $\underline{\mathcal{M}}(m, -)$ [checking required].

Definition 2.15. A $\mathcal{V}$-category $\underline{\mathcal{M}}$ is **cotensored**[a] if for each $v \in \mathcal{V}$ and $n \in \mathcal{M}$, there is an object $n^v \in \mathcal{M}$ with natural isomorphisms,

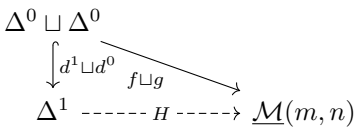
$$\underline{\mathcal{M}}(m, n^v) = \underline{\mathcal{V}}(v, \underline{\mathcal{M}}(m, n))$$

[a] there is no universally accepted definition of cotensored categories, I am just presenting the one given in Riehl's Book.

And with this, as one would expect, any closed symmetric monoidal category is tensored, cotensored and self-enriched. All self-enriched objects are also tensored and cotensored ,i.e.

the adjunction becomes an equivalence. We are now well equipped to discuss simplicial model categories and simplicially enriched categories in great detail as we shall soon see into the future that these class of examples are extremely important as any model category is "equivalent" to a simplicial model category in some sense, which we shall explore further on. It is essentially to say that any information given in a model category can be reinterpreted to uniquely (up to equivalence) create a simplicial model category. We shall now make all our previous intuitionistic statements precise.

If $\underline{\mathcal{M}}$ is a simplicially enriched category, then there is a canonical notion of homotopy between maps in the underlying category $\mathcal{M}$. A homotopy between $f, g :: m \to n$ is given by the following diagram.



Where H is the witness of the homotopy. If $\underline{\mathcal{M}}$ is also tensored, cotensored or closed monoidal then you may also create curried versions of the above diagram. The consequence of such a currying will be that the above diagram that is present in $\underline{\mathcal{V}}$ will then be present in the $\underline{\mathcal{M}}$ which may result in futher concrete criterions of when two maps are homotopic if there is a concrete realisation of the category $\underline{\mathcal{M}}$ itself and thus concrete calculations regarding homotopic diagrams which is precisely what will also help in calculating homotopy limits/colimits I believe. This also leads to a notion of homotopic equivalence which one can formalise easily. We now expand on how simplicially enriched tensored categories can further admit much more geometric characterisation than we initially realised.

Definition 2.16. If $\mathcal{M}$ is simplicially enriched, tensored, and cocomplete, the **geometric realisation** of a simplicial object is

$$|X_\bullet| = \int^{n \in \Delta} \Delta^n \otimes X_n.$$

This shall define a functor $|-| : \mathcal{M}^{\Delta^{op}} \to \mathcal{M}$.

We shall now leave this discussion to pursue homotopy (co)limits and how to compute them as we have not yet formally defined model categories and this treatment will be much more fruitful once we are well equipped on how to deal with derived categories.

2.3 Homotopy (Co)Limits

3 Weak Factorisation Systems and Model Categories

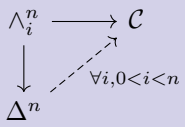
4 Infinity Categories

4.1 Introduction

We first deal with size issues to resolve paradoxes which occur without treating them, like the Russell's Paradox. Our solution to this is going to be Grothendiecks' Universes. A Grothendieck Universe $\mathcal{U}$ simply means that its subsets and elements constitute the model

for sets as we know them. More specifically, we call the elements of $\mathcal{U}$ "small sets". Thus when we take the set of all "small sets" still is a set but is no longer a "small set" but rather a "large set". The categories that we will not be a locally small categories which is precisely saying that the object and morphism set of the category are not necessarily, "small sets". Whenever we write **Set**, we mean the category of Sets. Now, we jump straight into the definition of an infinity category.

Definition 4.1. An ∞-category is a (large) simplicial set $\mathcal{C}$, such that it has the weak inner Kan Complex condition (the uniqueness is weakened to an existence excluding outer horns) ,i.e. there always exists an extension of the following map.[a]



[a]Please note that this is the definition of a (∞,1)-category for those who are familiar with the notation.

This is essentially the to preserve the infinity categorical data in terms of nerves of higher dimensional categories. Imagine it like this, we have the following data, objects, morphisms, homotopies between morphisms, homotopies between homotopies, ... what we want, is to encode this structure into a way such that we have a singular framework which we can fit any infinity category into. The solution we are using is to collect all objects into 0-simplices, all morphisms into 1-simplices, all homotopies into 2-simplices, all homotopies between homotopies into 3-simplices, ... this retains the information of homotopy as, if a particular simplex is filled then there is a homotopy that we can witness which makes 2 maps equivalent. Here, the composition of paths is just following the first 1-simplex and then the next one, which is not exactly equal to the composite, but there exists a homotopy between them which makes them equivalent[4]. In this light, the above definition should now make sense. Not only that but if you take $n = 1$ and both the zero-simplices to be the same object, then that gives us the existence of identities for each object (in similar fashion, we obtain the identities for any n-homotopy). Not that we do not require precisely the condition that identities should work as identites but rather we consider the extension itself to work the way identities are supposed to work for us.

Definition 4.2. A functor between ∞-categories is just a simplicial map.

Now, we shall give the definition of when exactly are 2 maps homotopic.

[4]This is precisely why the uniqueness statement is replaced by an existence statement purely because there is not a paramatrisation of composition of path, therefore what we mean to say is that any paramatrisation of the path that you pick are equivalent which required the Kan Horn Extension condition to weaken to an existence rather than a uniqueness because there may exist a ton of different paramatrisations which are essentially just equivalent.

Definition 4.3. Two maps f, g are homotopic if there exists 2-simplex s.t its 0th boundary is f, and 1st boundary is g, or vice versa while the last one is the respective identity. There is indeed a dual equivalent definition to this which can be used in various scenarios ,i.e. thre exists a 2-simplex s.t its 2nd boundary is f, and 1st boundary is g, or vice versa while the last one is the respective identity. Note that this defines an equivalence relation.

If one correctly understands the above definitions then they can accurately predict that the Singular Set of a Kan Complex (by singular set, I mean the simplicial set which generates the singular homology chain of any complex) is also an ∞-category but also an ∞-groupoid because the outer horn extension conditions will precisely give us the existence of the inverses for each map and homotopy.

Definition 4.4. A composition of two maps f, g is just the extension of a (2,1)-horn wherein 0th boundary is f and 2nd boundary is g.

I hope you saw this coming. Now we will go on to remind ourselves of the definition of the mapping space of 2 simplicial sets which is of great importance as it is exactly what a functor category is desired to be under some asssumptions.

Definition 4.5. Let $\mathcal{C}, \mathcal{D}$ be simplicial sets, then $Fun(\mathcal{C}, \mathcal{D})$ is also a simplicial set with $Fun(\mathcal{C}, \mathcal{D})_n$ = set of all simplicial maps from $\mathcal{C} \times \Delta^n \to \mathcal{D}$.

One can imagine the above definition to work in the form of maps from $\mathcal{C} \to \mathcal{D}$ be points and the natural transformations be edges and we are trying to give simplicial structure by collecting all Δ^n shaped collection of maps along with their respective natural transformations, ... into a set. The boundary maps are also the obvious boundary maps, just take the boundary of the Δ^n and retain information corresponding to that face and natural transformations, ... while forgetting the information corresponding to the rest of the faces. Now, we shall explore when exactly is this mapping space an infinity-category and the answer is ... when D is an infinity category.

Proposition 4.6. If $\mathcal{C}$ is a simplicial set and $\mathcal{D}$ is an infinity category. Then $Fun(\mathcal{C}, \mathcal{D})$ is also an infinity category.(rather it is an infinity groupoid)[a]

[a]For those who concern themselves with this more seriously, the functor category of 2 (∞,n)-categories is a (∞,n-1)-category. Please note that (in the case of (∞,1)-category) this is essentially just moving one level up, i.e. the morphisms are points, natural transformations are maps along with the homotopies, ... but when we say it is a groupoid, we do not mean that all natural transformations are a natural isomorphism, rather it just means that all homotopies which witness the commutativity of naturality squares in the natural transformation are invertible, which is just reflective of the fact that all homotopies are invertible in $\mathcal{D}$. This is a subtle point.

We now consider an equivalence between infinity categories.

Definition 4.7. $f : \mathcal{C} \to \mathcal{D}$ is an equivalence if there exists a $g : \mathcal{D} \to \mathcal{D}$ along with natural isomorphisms such that $fg \cong \mathbf{1}_{\mathcal{D}}$ and $gf \cong \mathbf{1}_{\mathcal{C}}$.

Now, we would like to develop the alternative characterisation of an equivalence via a functor being essentially surjective and fully faithful, but that is not as easy to implement because while our intuition is right in the sense of fixing an equivalent object, morphism, homotopy, ... to its corresponding object, morphism, homotopy, ... in our source category. We must first fix this into a known framework of simlicial set first instead of giving infinite number of criterions in our definition. Hence, we will have to discuss what precisely is the mapping space between 2 objects in an infinity category.

Definition 4.8. Let $a, b \in \mathcal{C}$, we define the mapping space of all functions from a to b to be the following pullback.

$$\begin{array}{ccc} Map_{\mathcal{C}}(a,b) & \longrightarrow & Fun(\Delta^1, \mathcal{C}) \\ \downarrow & & \downarrow{\scriptstyle f \mapsto (\partial_0(f), \partial_1(f))} \\ \Delta^0 & \xrightarrow[\star \mapsto (a,b)]{} & \mathcal{C} \times \mathcal{C} \end{array}$$

One can easily notice the simplicial structure on $Map_{\mathcal{C}}(a, b)$ which is basically just points being maps from a to b, homotopies between maps being 1-simplicies, homotopies between homotopies being 2-simplices, ... but if $\mathcal{C}$ is just an infinity category then $Map_{\mathcal{C}}(a, b)$ is just an infinity groupoid because all homotopies in $\mathcal{C}$ are invertible ,i.e. it is a Kan Complex. Another simple but powerful observation that they make in the lecture is that $\pi_0(Map_{\mathcal{C}}(a, b))$ is precisely the equivalence classes of maps under the homotopic equivalence relation[5]. Now, we are capable to define what exactly are fully faithful infinity functors.

Definition 4.9. $f : \mathcal{C} \to \mathcal{D}$ is **Fully Faithful** if $Map_{\mathcal{C}}(a, b) \to Map_{\mathcal{D}}(F(a), F(b))$ are homotopy equivalent for all $a, b \in \mathcal{C}$.

Definition 4.10. $f : \mathcal{C} \to \mathcal{D}$ is **Essentially Surjective** if for every $d \in \mathcal{D}$, there exists a $c \in \mathcal{C}$ such that $d \cong F(c)$.

When both of the above criterions are true, what we get is a a criterion of when exactly is a functor an equivalence[6].

Proposition 4.11. A functor is an equivalence if and only if it is **Fully Faithful** and **Essentially Surjective**.

[5]This is an equivalence relation only because $\mathcal{C}$ is an $(\infty, 1)$-category, i.e. all homotopies are invertible.

[6]Thinking about equivalence, a person might think of how does one formulate adjunctions in the language of infinity categories and although we might see this ahead, I believe it is just a pair of opposite facing functors and using the initial object in the comma category to get the units and counits of the adjunction. This essentially solves the same problem as before that we would like to not have infinite number of criterions in our definitions due to the existence of higher homotopies.

We shall now define what are subcategories of infinity categories and it is essentially the same notion as regular category theory of closing a subset of objects and maps under compositions.

Definition 4.12. Given $S \subset \mathcal{C}_0$ then we define $\mathcal{C}_S \subset \mathcal{C}$ consisting of all simplicies with all vertices in S. This will result in an infinity category.

We now try to revisit the idea of composition in the light of the definition of mapping spaces and what we shall realise is that composition is a mp between mapping spaces. We had before realised that composition of 2 maps is essentially just a 2-horn being filled which gives us the 2-simplex. But, the mapping spaces are filled with points which correspond to maps. Thus, we really want to paramatrise each 2-simplex in a way just that it becomes an element of a mapping space such that we are able to compose 2 mapping spaces. We can bring this intuition to life with the following definition.

Definition 4.13. Fix any $a, b, c \in \mathcal{C}$, the following pullback is the composition mapping space of a, b, c.

$$\begin{array}{ccc} Map_{\mathcal{C}}(a,b,c) & \longrightarrow & Fun(\Delta^2, \mathcal{C}) \\ \downarrow & & \downarrow{\scriptstyle f \mapsto (f(0), f(1), f(2))} \\ \Delta^0 & \xrightarrow[\star \mapsto (a,b,c)]{} & \mathcal{C} \times \mathcal{C} \times \mathcal{C} \end{array}$$

Proposition 4.14. $Map_{\mathcal{C}}(a, b, c)$ is a Kan Complex(aka Infinity Groupoid) and $\partial_1(Map_{\mathcal{C}}(a, b, c)) = Map_{\mathcal{C}}(a, c)$. This should hint towards the other restriction map as it is a direct paramatrisation of 2-simplices made by horn filling of composable maps. Although a very important point is that in this light, composition itself becomes a map by considering the inverse of the left map in the following diagram[a].

$$Map_{\mathcal{C}}(a,b) \times Map_{\mathcal{C}}(b,c) \xleftarrow{\sim} Map_{\mathcal{C}}(a,b,c) \xrightarrow{\partial_1} Map_{\mathcal{C}}(a,c)$$

[a] This could have been guessed with a closer look into how composition works in infinity categories :P, silly me.

Now, we shall try to work with the "category of spaces" wherein the spaces for us are precisely Kan Complexes. The reasoning of why we call this category of spaces is simply that we know that any topological space can be reduced to a CW-Complex as far as the information of homotopy groups is concerned but the structure of a CW-Complex itself can be reduced to a Simplicial Set structure using the Singular Set Functor (the functor on which the free abelian groups are generated to create the Singular Chain Functor). This will result in a Kan Complex always. Hence, studying the category of all Kan Complexes is equivalent to studying CW-Complexes and thus homotopy groups of any Topological Space. The category of spaces is also going to be an infinity category ,i.e. it will be enriched over simplicial sets which are going to be infinity groupoids because all homotopies are going to be invertible, note that this is just a shadow of the structure of homotopy and homotopy equivalences derived from the category of Topological Spaces. It is in this precise way that

the "category of spaces" is going to be an infinity category because we know the category of topological spaces itself is an infinity category. We call this category spaces, **Kan**.

Definition 4.15. Simplicial Set S. Define $S_n = Fun(\Delta^n, \mathbf{Kan})$ i.e. 0-simplices, Kan Complexes and 1-simplices, Maps between Kan Complexes and 2-simplices, Homotopies between Maps and ..., this makes S into an infinity category which is of our main interest.

We shall now move onto how does even work with (co)limits in the infinity categories.

4.2 Infinity Categorical (Co)Limits

We all know from 1-category that a (co)limit is just a universal (co)cone over some diagram $\mathcal{D}$. We are going to exactly analogously construct the cone over some simplicial set (analogous to a diagram in 1-category) in an infinity category and then define the universal property as the initial or terminal object in the category of cones over a simplicial set. This should be natural because every diagram not only has the objects, maps, but also homotopies and homotopies between homotopies, ... which can all be framed into a single simplicial set. From here onwards, I will be refering to all homotopies and higher homotopies as "higher coherence data" as it is the usual convention and expect the reader to adapt to the language aswell. Thus, our first definition of a cone arrives as follows.

Definition 4.16. A **cone** over $F : D \to \mathcal{C}$ is a pair consisting of an object $y \in \mathcal{C}$ and a natural transformation from $\eta : 1_y \Rightarrow F$. Here, D is a simplicial set, $\mathcal{C}$ an infinity category, the natural transformation is a 1-morphism in the infinity category $Map(I, \mathcal{C})$.[a]

[a] One might wonder where the objects $I, \mathcal{C}$ live ? They are in an $(\infty, 2) - category$, i.e. a category that contains all infinity categories.

One can dually define the cocone which is also useful the construction of colimits. Theres another way to create cones which is to consider the join construction $(\star \wedge I)$ and then consider all functors from the join to our category wherein the image of $\star$ will always be the cone. This is essentially the same consideration instead you are making the cone, a part of the diagram data now and there is no other difference. One may also see the dual of this construction gives the cocone.

Definition 4.17. One can check that there exists an infinity category of cones, which is true because of the functor category from the join construction.
A cone (y, η) is a **limit cone** if for all $x \in Fun((\star \wedge I), \mathcal{C})$, the kan complex $Map_{\mathcal{C}}(y, x)$ is contractible.
Equivalently, the canonical map of precomposition $Map_{\mathcal{C}}(x, y) \to Map_{\mathcal{C}^{(\star \wedge \mathcal{C})}}(1_x, F)$ is a homotopy equivalence.

One can now realise that limits are unique upto canonical equivalence. Thus the category of limit cones over a functor is a contractible space.

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